



Team Details

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- Problem Statement No.: 004
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Idea Title (Proposed Solution)

- **The Problem:** Academic collaboration is fragmented across classrooms, WhatsApp/Telegram groups, Discord, LinkedIn, GitHub, and research platforms. Students often hesitate to ask questions, struggle to find the right teammates or mentors, and have no single profile to showcase their real academic contributions.
- **Solution:** Connext is a unified academic collaboration platform that connects students, faculty, researchers, and mentors beyond their individual campuses.
- **How it Works (3 Pillars):**
 - **Academic Passport:** Combines education, skills, projects, achievements, GitHub, research, and other verified profiles into one academic identity.
 - **Smart Collaboration:** AI matches students with relevant teammates, mentors, projects, research, and communities based on skills, interests, and requirements.
 - **Contribution & Community:** Q&A, cross-campus guilds, project collaboration, XP, streaks, quests, and reputation turn learning into measurable contribution.
- **Impact:** Breaks campus boundaries and makes it easier to ask questions, find collaborators, showcase real work, and access mentorship and opportunities.
- **Innovation:** A Proof-of-Contribution Graph connects a user's skills → projects → contributions → repositories → research → achievements, creating a profile based on evidence rather than popularity.



Technical Approach

Technologies to be Used:

- **Frontend:** Next.js, React, TypeScript, Tailwind CSS
- **Backend:** Node.js, FastAPI, REST APIs
- **Database:** PostgreSQL + Redis
- **AI:** LLMs, embeddings, Vector Database for semantic search & matching
- **Integrations:** GitHub, LinkedIn, Google, ORCID, ResearchGate
- **Deployment:** Docker, Cloudflare, GitHub Actions

Prototype Flow:

User → Academic Passport → AI Matching → Discover → Collaborate →
Verify Contribution → XP/Reputation → Opportunities

Security & Scalability:

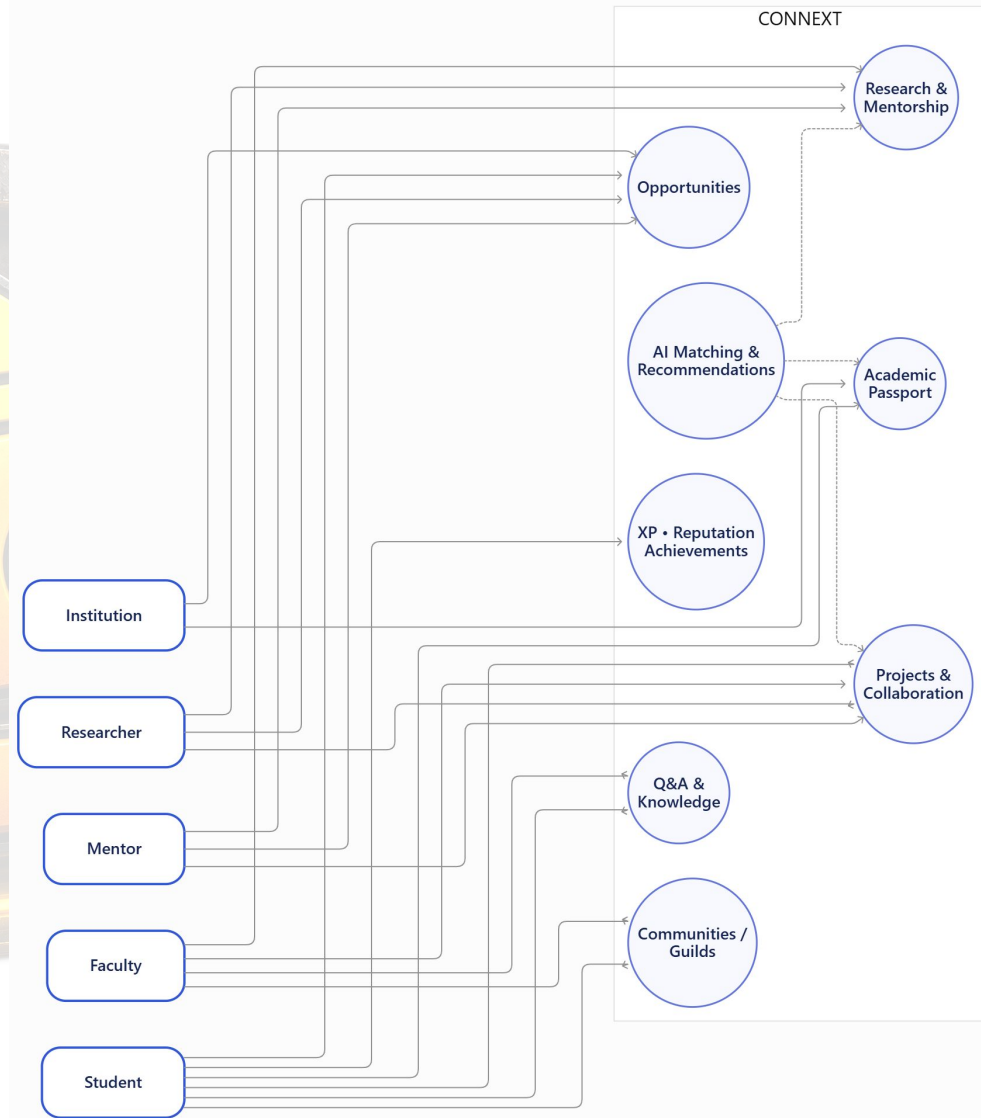
- OAuth-based authentication, role-based access, privacy controls, API security, caching and modular cloud deployment.

Methodology & Process:

1. **User Onboarding:** Register → Verify Institution → Create Academic Passport
2. **Profile Integration:** Connect GitHub / LinkedIn / Research profiles → Build Skill Graph
3. **Discovery:** AI recommends questions, people, projects, mentors and communities
4. **Collaboration:** Create/join projects → communicate → contribute → track milestones
5. **Contribution Engine:** Verify contributions → award XP, reputation, streaks and achievements
6. **Growth:** Update Academic Passport → discover research, mentorship and career opportunities



Use Case





Feasibility and Viability

Feasibility

- **Technically Feasible:** Uses scalable and widely adopted web, cloud, database, and AI technologies.
- **Low-Cost MVP:** Can be developed incrementally with open-source tools and APIs.
- **Scalable:** Modular architecture supports expansion from colleges to a nationwide network.

Challenges & Risks

- **User Adoption:** Building an active community across institutions.
- **Privacy & Security:** Protecting academic profiles and connected accounts.
- **Fake Contributions:** Risk of false projects, achievements, or credentials.
- **AI Accuracy:** Incorrect recommendations or generated information.

Strategies

- **Verified Identity:** Institution verification + OAuth-based integrations.
- **Moderation:** Reporting, reputation systems, and AI-assisted content moderation.
- **Evidence-Based Profiles:** Link projects, repositories, publications, and achievements as proof.
- **Phased Rollout:** Start with a few institutions, validate, then scale nationally.



Research and References

- APAAR — One Nation, One Student ID <https://apaar.education.gov.in/about>
- Stack Overflow — Reputation System <https://stackoverflow.com/help/whats-reputation>
- GitHub — Pull Requests & Collaboration <https://docs.github.com/en/pull-requests/about-pull-requests>
- LinkedIn — Profile / Academic & Professional Identity <https://www.linkedin.com>
- ResearchGate — Research Collaboration Network <https://www.researchgate.net/about>
- ORCID — Persistent Researcher Identity <https://orcid.org/>
- Student Question-Participation Research — PLOS ONE Call on me! Undergraduates' perceptions of voluntarily asking and answering questions in front of large-enrollment science classes <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0243731>
- Student Team Formation Research — Learning and Instruction Antecedents of student team formation in higher education <https://doi.org/10.1016/j.learninstruc.2024.101931>
- Gamification & Student Engagement — Systematic Review Impact of gamification on school engagement: a systematic review <https://doi.org/10.3389/feduc.2024.1466926>
- Discord — Community & Communication Platform <https://discord.com/safety>